

AI Intelligent Freight Quote Generation System

Milestone 3 — Complete Technical Scope, Models, Schemas, APIs, Documents & Full Project Flow

Purpose: This document converts the M3 requirements into an implementation-ready blueprint. It covers Weather Intelligence, Customs Intelligence, Shipment Risk Scoring, ML Pricing, database models, API contracts, frontend modules, documentation, testing and the complete end-to-end project flow.

Important scope note: The project materials describe M3 in two ways: the detailed technical blueprint defines M3 as Weather Risk & Customs Intelligence, while the kickoff material also expects a real ML pricing model and a rule-vs-ML comparison. This document therefore includes both so no milestone deliverable is missed.

1. Full Project Flow

The complete system starts with shipment input and moves through routing, pricing, weather intelligence, customs intelligence and risk aggregation before producing a quote. After M3, risk and compliance become part of the quote decision rather than separate reports.

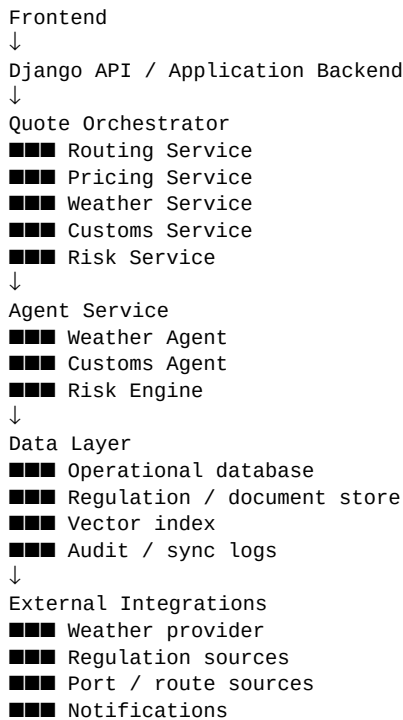
```
Customer / Sales User
↓
Create Shipment
↓
Validate Shipment Data
↓
Route Intelligence
  ■■■ Candidate Routes
  ■■■ Distance / ETA
  ■■■ Ports / Chokepoints
  ■■■ Route Risk
↓
Parallel Intelligence
  ■■■ Pricing Engine
  ■ ■■■ Rule-based pricing
  ■ ■■■ ML price prediction
  ■■■ Weather Agent
  ■ ■■■ Route geometry sampling
  ■ ■■■ Weather observations
  ■ ■■■ Storm detection
  ■ ■■■ Delay probability
  ■■■ Customs Agent
  ■■■ HS code / commodity validation
  ■■■ Regulation retrieval
  ■■■ Required documents
  ■■■ Compliance risk
↓
Risk Engine
  ■■■ Weather score
  ■■■ Customs score
  ■■■ Route score
  ■■■ Port / chokepoint score
  ■■■ Cargo risk
↓
Composite Shipment Risk
↓
Explainability + Alerts
↓
Customs Sign-off when required
↓
Quote Decision
  ■■■ Approved → Quote issued
  ■■■ Needs review → Human action
  ■■■ Blocked → Cannot issue quote
↓
Quote Detail / Customer Delivery
↓
Audit + Monitoring + Feedback
```

2. Milestone Roadmap

Milestone	Main responsibility	Output
M1	Foundation + route intelligence	Shipment, route graph, candidate routes, route API
M2	Pricing + quote generation	Rule pricing, quote builder, quote APIs
M3	Weather + Customs + Risk + ML pricing	Risk/compliance intelligence and ML comparison
M4	Production hardening	Monitoring, reliability, security, performance, deployment

M3 is the intelligence and compliance layer that sits between routing/pricing and final quote issuance.

3. M3 Architecture



4. M3 Functional Modules

Weather Intelligence: assess weather along active routes, identify storms and severe conditions, calculate weather risk, estimate delay probability, and provide alternative-route advice.

Customs Intelligence: validate origin/destination, HS code, commodity and Incoterm; retrieve applicable regulations; determine required documents; flag compliance risks; produce a readiness score; support human sign-off.

Shipment Risk Engine: combine weather, customs, route, port/chokepoint and cargo factors into an explainable composite risk score.

ML Pricing: train a freight-price model, evaluate it with MAE/RMSE/R², and compare ML predictions against the existing rule-based formula.

5. Database Models & Schemas

WeatherAssessment

id, shipment_id, quote_id, route_id, risk_score, risk_level, storm_risk, rainfall_risk, wind_risk, wave_risk, temperature_risk, delay_probability, assessment_status, provider, provider_timestamp, assessed_at, expires_at, confidence_score, created_at, updated_at

WeatherObservation

id, route_id, latitude, longitude, observation_time, temperature, wind_speed, wind_direction, rainfall, wave_height, visibility, pressure, weather_condition, storm_detected, storm_type, storm_severity, provider, raw_payload, created_at

WeatherAlert

id, shipment_id, route_id, alert_type, severity, title, message, latitude, longitude, starts_at, ends_at, status, acknowledged_at, resolved_at, created_at

CustomsRequirement

id, origin_country, destination_country, hs_code, commodity, incoterm, requirement_type, description, mandatory, risk_level, regulation_id, effective_from, effective_to, active, created_at, updated_at

CustomsDocumentRequirement

id, customs_requirement_id, document_type, document_name, mandatory, description, accepted_formats, created_at

CustomsComplianceCheck

id, shipment_id, quote_id, origin_country, destination_country, hs_code, commodity, incoterm, readiness_score, risk_level, status, checked_at, expires_at, created_by, reviewed_by, created_at, updated_at

CustomsChecklistItem

id, compliance_check_id, requirement_id, item_name, description, mandatory, status, document_required, document_uploaded, evidence, citation, reviewer_comment, created_at, updated_at

RegulationDocument

id, title, country, authority, document_type, source_url, source_name, version, effective_from, effective_to, published_at, content, status, last_synced_at, created_at, updated_at

RegulationChunk

id, regulation_document_id, chunk_index, content, embedding, metadata, page_number, section_name, effective_from, effective_to, created_at

HSCodeReference

id, hs_code, description, chapter, heading, subheading, commodity_type, restricted, prohibited, country, effective_from, effective_to, source, created_at, updated_at

ShipmentRiskAssessment

id, shipment_id, quote_id, weather_score, customs_score, route_score, port_score, overall_score, risk_level, confidence_score, explanation, assessed_at, model_version, created_at, updated_at

RiskFactor

id, risk_assessment_id, factor_type, factor_name, score, weight, contribution, severity, reason, source, created_at

RiskAlert

id, shipment_id, quote_id, risk_assessment_id, alert_type, severity, title, message, source, status, acknowledged_by, acknowledged_at, resolved_by, resolved_at, created_at

ShipmentDocument

id, shipment_id, customs_check_id, document_type, file_name, file_url, mime_type, file_size, uploaded_by, uploaded_at, verification_status, verified_by, verified_at, rejection_reason

DataFreshness

id, provider, data_type, last_success_at, last_attempt_at, record_count, status, freshness_seconds, error_message, created_at, updated_at

IntegrationSyncLog

id, provider, integration_type, started_at, completed_at, status, records_processed, records_failed, error_message, request_id, created_at

AlertSubscription

id, user_id, alert_type, severity, email_enabled, teams_enabled, slack_enabled, active, created_at, updated_at

6. Core Relationships

Customer → Shipment → Route
Shipment → Quote
Route → WeatherAssessment → WeatherObservation
Shipment → CustomsComplianceCheck → CustomsChecklistItem
CustomsRequirement → RegulationDocument → RegulationChunk
Shipment/Quote → ShipmentRiskAssessment → RiskFactor
Shipment/Quote → RiskAlert
Shipment → ShipmentDocument
Provider → IntegrationSyncLog / DataFreshness

7. API Contracts

Method	Endpoint	Purpose
POST	/v1/weather/assess	Assess weather risk for route and departure window
POST	/v1/customs/validate	Validate customs requirements and documentation
POST	/v1/regulations/search	Search regulation corpus with citations
GET	/v1/risk/{shipment_id}	Get current composite shipment risk
POST	/v1/risk/assess	Create/recalculate risk assessment
POST	/v1/customs/{check_id}/sign-off	Approve/reject/conditionally approve customs
GET	/v1/alerts	List alerts
POST	/v1/alerts/{id}/acknowledge	Acknowledge alert

8. Weather Agent Flow

Shipment Route
↓
Route geometry
↓
Sample points along route
↓
Weather provider request
↓
Normalize observations
↓
Detect storm / severe conditions
↓
Calculate weather risk
↓
Predict delay probability
↓
Generate alternative route advice
↓
Persist WeatherAssessment + Observations + Alerts
↓
Return explainable result

Suggested delay model inputs: wind speed, rainfall, wave height, visibility, storm distance, storm severity, route length, departure month, port congestion and related route conditions.

9. Customs Agent + RAG Flow

Shipment details
↓
Origin + Destination + HS Code + Commodity + Incoterm
↓
Validate basic fields
↓

```

Retrieve regulation documents
↓
Chunk + embeddings
↓
Hybrid search (keyword + vector)
↓
Reranking
↓
Extract applicable requirements
↓
Determine mandatory documents
↓
Check restrictions / prohibitions
↓
Generate citations/evidence
↓
Calculate customs readiness
↓
Create checklist
↓
Human compliance sign-off
↓
PASS / FAIL / NEEDS_DOCUMENTS / NEEDS_REVIEW

```

10. Risk Engine

A configurable weighted score is recommended for the first version. Example weights:

```

Weather 30%
Customs 25%
Route 20%
Port 15%
Cargo 10%

```

```

overall_score =
weather_score × 0.30
+ customs_score × 0.25
+ route_score × 0.20
+ port_score × 0.15
+ cargo_score × 0.10

```

Example thresholds:

```

0-30 LOW
31-60 MEDIUM
61-80 HIGH
81-100 CRITICAL

```

Every score should have factor-level explanation: factor, score, weight, contribution, severity, reason and source.

11. Quote State Machine Integration

```

Quote Draft
↓
Route Validated
↓
Pricing Calculated
↓
Weather Assessed
↓
Customs Assessed
↓
Risk Calculated
↓
Is customs sign-off required?
■■■ NO → Continue
■■■ YES
↓
Compliance Review
■■■ REJECTED → BLOCK QUOTE
■■■ NEEDS_DOCUMENTS → HOLD
■■■ APPROVED → Continue

```

↓
Risk Threshold Check
■■■ Critical → Human Review / Block according to policy
■■■ Acceptable → Continue
↓
Quote Issued
↓
Audit Record

12. ML Pricing Deliverable

Historical freight data
↓
Cleaning
↓
Feature engineering
↓
Train/test split
↓
Baseline model
↓
Candidate regression models
↓
Evaluation
■■■ MAE
■■■ RMSE
■■■ R²
↓
Select best model
↓
Save model + preprocessing artifacts
↓
Prediction API
↓
Compare:
Rule Price vs ML Price
↓
Document recommendation

Suggested features: distance, weight, volume, cargo type, transport mode, origin, destination, season/month, fuel price, carrier, container type and other available historical variables. Target: actual freight price.

13. Repository Structure

```
backend/  
■■■ apps/  
■ ■■■ shipments/  
■ ■■■ quotes/  
■ ■■■ routes/  
■ ■■■ pricing/  
■ ■■■ risk/  
■ ■■■ customs/  
■ ■■■ integrations/  
■■■ agents/  
■ ■■■ weather/  
■ ■■■ customs/  
■ ■■■ risk/  
■■■ documents/  
■■■ notifications/  
■■■ audit/  
  
agent-service/  
■■■ agents/  
■ ■■■ weather_agent/  
■ ■■■ customs_agent/  
■ ■■■ risk_engine/  
■■■ api/  
■■■ integrations/  
■■■ models/  
■■■ evaluation/
```

```
ml/  
■■■ data/  
■■■ notebooks/  
■■■ src/  
■■■ models/  
■■■ reports/
```

14. Frontend Screens

1. Shipment Create
2. Route Selection / Route Map
3. Quote Builder
4. Weather Risk Panel
5. Customs Check
6. Customs Document Upload
7. Customs Officer Sign-off
8. Risk Dashboard
9. Alerts Inbox
10. Customs Dashboard
11. Regulation Library
12. Quote Detail with risk explanation
13. ML vs Rule Price comparison (internal/admin)
14. Integration / data freshness monitoring

15. Documents You Must Produce

- M3_Requirements.md
- M3_Architecture.md
- M3_Database_Design.md
- M3_API_Documentation.md
- Weather_Agent_Design.md
- Customs_Agent_Design.md
- Shipment_Risk_Scoring.md
- RAG_Design.md
- ML_Model_Evaluation.md
- Dataset_Documentation.md
- M3_Test_Plan.md
- M3_Test_Results.md
- M3_Definition_of_Done.md
- ADRs for major architecture decisions

16. Testing & Acceptance

Functional tests

- Weather assessment for every active route
- Customs checklist generation
- Required-document detection
- Regulation citation validation
- Risk calculation
- Alert generation
- Customs sign-off
- Quote blocking

Integration tests

- Weather provider failure / fallback
- Regulation source failure / stale data
- Database persistence
- Agent-to-backend contract
- Quote state transitions

ML tests

- Data leakage check
- Train/test evaluation
- MAE / RMSE / R^2
- Rule-vs-ML comparison

Target acceptance

- 100% of active routes receive weather risk
- $\geq 90\%$ correct customs checklists across 50 curated scenarios
- Risk score is explainable
- Required regulatory claims have evidence/citations
- Customs sign-off can block quote issuance

17. M3 Implementation Order

Phase 1 – Database & Contracts

1. Create migrations/models
2. Define serializers/schemas
3. Publish API contracts
4. Add indexes and audit fields

Phase 2 – Weather

5. Weather provider adapter
6. Route sampling
7. Weather observation persistence
8. Weather scoring
9. Alerts
10. Weather UI

Phase 3 – Customs

11. Regulation ingestion
12. Chunking/embeddings
13. Hybrid retrieval
14. Reranking
15. Customs validation
16. Checklist generation
17. Document upload
18. Sign-off workflow
19. Customs UI

Phase 4 – Risk

20. Risk factor framework
21. Composite scoring
22. Explainability
23. Risk alerts
24. Quote integration
25. Risk dashboard

Phase 5 – ML Pricing

26. Dataset preparation
27. Feature engineering
28. Train candidate models
29. Evaluate
30. Save model artifacts
31. Prediction endpoint
32. Rule-vs-ML comparison

Phase 6 – Testing & Delivery

33. Unit tests
34. Integration tests
35. 50 customs scenarios
36. Weather coverage test
37. E2E quote tests
38. Documentation
39. Demo / acceptance
40. M3 sign-off

18. Final End-to-End Example

Example shipment:

Origin: Chennai, India

Destination: Rotterdam, Netherlands

Commodity: Electrical equipment

HS Code: 850440

Incoterm: CIF

1. User creates shipment.
2. Routing service generates candidate route.
3. Pricing service calculates rule price and ML prediction.
4. Weather Agent samples route and finds severe weather near a route segment.
5. Weather risk = HIGH; delay probability = 68%.
6. Customs Agent validates HS code and retrieves Netherlands import requirements.
7. Customs checklist identifies mandatory documents.
8. Regulation evidence is attached to each important requirement.
9. Customs readiness = 88%.
10. Risk Engine combines weather, customs, route and port scores.
11. Composite risk = HIGH.
12. System explains: weather contributed most to the risk.
13. If policy requires customs sign-off, quote remains blocked until approval.
14. Officer reviews documents and signs off.
15. Quote becomes eligible for issuance.
16. Quote is issued with price, ETA, risk level, weather warnings and compliance status.
17. Audit trail stores every important decision.

19. M3 Definition of Done

- ✓ Weather Agent implemented
- ✓ Weather observations stored
- ✓ Weather risk calculated
- ✓ Delay probability available
- ✓ Storm alerts available
- ✓ Customs Agent implemented
- ✓ Regulation corpus available
- ✓ Hybrid RAG retrieval implemented
- ✓ Citations/evidence enforced
- ✓ Customs checklist generated
- ✓ Document upload implemented
- ✓ Compliance sign-off implemented
- ✓ Quote blocking implemented
- ✓ Composite risk score implemented
- ✓ Factor-level explainability implemented
- ✓ Alerts implemented
- ✓ Risk dashboard implemented
- ✓ Customs dashboard implemented
- ✓ ML pricing model trained/evaluated
- ✓ Rule-vs-ML comparison documented
- ✓ API contracts published
- ✓ Database migrations complete
- ✓ Tests complete
- ✓ Acceptance criteria demonstrated
- ✓ M3 documentation complete

Final architecture principle: M3 is not a collection of isolated features. Weather, Customs, Pricing and Risk must feed the same quote orchestration flow, with evidence, human approval and auditability built into the decision path.